

Predictive Modeling and Risk Scoring for Bank Customer Churn

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Abstract

Customer churn is an important business problem in banking because customer loss can affect recurring revenue, customer lifetime value, and future engagement opportunities. This paper presents a machine-learning framework for predicting bank customer churn and converting model probabilities into actionable risk scores. The project uses a structured customer dataset containing demographic, account, product, activity, balance, and salary-related attributes, with Exited as the churn target. The workflow includes data preparation, exploratory analysis, feature engineering, supervised classification, model evaluation, risk-band generation, and deployment through a Streamlit web application. Logistic Regression, Random Forest, Gradient Boosting, and XGBoost were evaluated as candidate classifiers. The deployed application accepts customer information and returns a churn probability and risk category. The study also discusses deployment compatibility, reproducibility, limitations, and responsible use of predictive scores.

Keywords: Customer Churn, Banking Analytics, Machine Learning, Risk Scoring, Classification, Predictive Modeling, XGBoost, Random Forest, Streamlit

1. Introduction

Customer retention is a significant analytical use case in banking. Banks serve customers through deposits, cards, loans, digital services, and other financial products. When a customer leaves, the organization may lose current revenue and future business opportunities. Because customer records contain behavioral and financial signals, machine learning can estimate the probability that a customer will churn. This project develops an end-to-end classification workflow that produces a probability-based risk score and maps it to practical risk bands. The system is intended as decision support rather than an automatic decision maker.

2. Problem Statement

Traditional customer analysis can identify historical churn patterns, but it may not provide a scalable method for estimating churn risk for individual customers. The problem addressed in this study is to build a supervised machine-learning system that learns from historical customer records, predicts churn probability for new records, and communicates the result through an accessible dashboard.

3. Objectives

- Develop a reproducible machine-learning pipeline for bank customer churn prediction.
- Explore demographic, account, product, financial, and engagement characteristics associated with churn.
- Engineer derived variables that capture relationships between customer attributes.
- Compare multiple classification algorithms using standard evaluation metrics.
- Convert predicted probabilities into interpretable customer risk bands.
- Deploy the trained model through a Streamlit application for interactive prediction and scenario analysis.
- Document technical limitations and deployment considerations.

4. Dataset and Variables

The project uses a bank customer dataset with 10,000 customer records and 14 original variables. The target variable is Exited, indicating whether the customer left the bank. Variables include CustomerId, Surname, CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember,

EstimatedSalary, and Exited. CustomerId and Surname are treated primarily as identifiers rather than predictive business variables.

Variable group	Examples	Role
Demographic	Age, Gender, Geography	Segmentation
Credit/account	CreditScore, Tenure, Balance	Financial/relationship context
Product/activity	NumOfProducts, HasCrCard, IsActiveMember	Usage and engagement
Income	EstimatedSalary	Financial context
Target	Exited	Churn outcome

5. Data Preprocessing and Exploratory Analysis

The preprocessing stage prepares the structured data for classification. Categorical variables are encoded for machine-learning algorithms, while numerical variables are handled within the modeling pipeline. Exploratory analysis examines target distribution, demographics, account characteristics, and engagement variables. The documented dataset contains approximately 20.37% exited customers and 79.63% retained customers, making precision, recall, F1-score, and ROC-AUC useful complements to accuracy.

6. Feature Engineering

Four derived features were incorporated: $\text{BalanceSalaryRatio} = \text{Balance}/(\text{EstimatedSalary}+1)$, $\text{ProductDensity} = \text{NumOfProducts}/(\text{Tenure}+1)$, $\text{EngagementProduct} = \text{IsActiveMember} \times \text{NumOfProducts}$, and $\text{AgeTenureInteraction} = \text{Age} \times (\text{Tenure}+1)$. These variables represent relative financial position, product intensity, engagement, and demographic-tenure interaction.

7. Machine-Learning Methodology

The project evaluates Logistic Regression as a linear baseline, Random Forest for nonlinear ensemble learning, Gradient Boosting for additive tree-based learning, and XGBoost as an optimized gradient-boosting implementation. XGBoost provides an `XGBClassifier` interface for binary classification. The workflow is data loading, validation, exploration, feature engineering, preprocessing, train/test splitting, model training, evaluation, probability-based risk scoring, model persistence, and deployment.

The trained model is persisted with joblib. Scikit-learn documentation notes that joblib can efficiently persist fitted estimators but that persisted models should be loaded with compatible Python and dependency versions; loading models across different scikit-learn versions is unsupported.

8. Model Evaluation

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	71.0%	38.3%	69.8%	49.4%	77.6%
Random Forest	86.7%	73.9%	53.6%	62.1%	85.9%
Gradient Boosting	87.1%	78.9%	49.6%	60.9%	87.0%
XGBoost	86.8%	78.0%	48.9%	60.1%	86.1%

The figures above are the performance values documented for this project and should be interpreted in the context of its dataset, split, preprocessing, and evaluation procedure. They do not establish equivalent performance on another bank or future population.

9. Churn Probability and Risk Scoring

The application uses predicted probability rather than only a binary label. The documented risk bands are Low (<0.30), Medium ($0.30\text{--}<0.60$), High ($0.60\text{--}<0.80$), and Critical (≥ 0.80). These are project-defined operational

categories, not universal banking standards. Thresholds should be validated against the costs of missed churn and unnecessary retention outreach.

10. Streamlit Application

The trained model is integrated into a Streamlit application. The app loads the model bundle, accepts customer attributes, performs the required preprocessing, generates a churn probability, and displays the risk category. The source code and model artifact are maintained in GitHub and the application is deployed through Streamlit Community Cloud. The deployment environment was aligned with the model-training environment because persisted scikit-learn models require compatible dependencies. ■cite■turn0search2■

11. Business Use and Interpretability

The system is intended to support retention analysis. A churn probability can help prioritize customers for human review, targeted communication, personalized offers, or additional service analysis. Predictions should be combined with current customer information and business rules before operational action.

12. Limitations

- Historical data may not represent future customer behavior or changing economic conditions.
- Performance on the project dataset does not guarantee equivalent performance for another bank or population.
- Probability scores are estimates rather than certainties.
- Risk thresholds are project-defined and should be validated against business costs.
- Joblib/pickle artifacts should only be loaded from trusted sources and with compatible environments. ■cite■turn0search0■turn0search2■
- Production use would require monitoring, drift detection, calibration, fairness analysis, and periodic retraining.

13. Responsible and Ethical Use

Customer churn prediction should be used as decision support rather than as an automatic basis for adverse treatment. Relevant customer groups should be evaluated for performance differences, and potentially sensitive or proxy variables should be reviewed carefully. Human oversight, data-protection controls, model monitoring, and documented governance are recommended for consequential use.

14. Conclusion

This study presents an end-to-end bank customer churn prediction and risk-scoring system combining structured data analysis, feature engineering, supervised classification, probability-based risk categorization, model persistence, and interactive deployment. The project demonstrates how historical customer information can be transformed into a practical analytical tool for identifying potential churn risk. Future work can extend the system with calibrated probabilities, explainability, automated monitoring, drift detection, cost-sensitive optimization, and secure model serving.

15. Future Enhancements

- Add SHAP or other model-explanation methods for customer-level explanations.
- Perform probability calibration and cost-based threshold selection.
- Introduce data-drift and model-performance monitoring.
- Evaluate subgroup fairness before operational use.
- Automate periodic retraining with versioned datasets and reproducible environments.

- Consider interoperable model formats or containerized deployment where appropriate.

16. References

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Project Repository: <https://github.com/Greeshma-stack/bank-customer-churn-prediction>

Deployment: Streamlit Community Cloud